

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 2: OBJECT LABELING AND COUNTING

SLO 1: Identify and label individual objects in a binary image using connected component analysis.

MULTIPLE CHOICE QUESTIONS SOLUTIONS

1. Connected component labeling is used to:

☒ **Identify separate objects in an image**

2. Which connectivity considers horizontal and vertical neighbors only?

☒ **4-connectivity**

3. Which connectivity considers diagonal neighbors as well?

☒ **8-connectivity**

OBJECT LABELING ACTIVITY SOLUTIONS

Binary Matrix Input:

```
0 0 0 0 0
0 1 1 0 1
0 1 1 0 1
0 0 0 0 0
```

1. How many separate objects are present in the image?

2 separate objects (Object 1: 2x2 block on the left; Object 2: 2x1 block on the right).

2. Label the objects using numbers (1, 2, 3...):

```
0 0 0 0 0
0 1 1 0 2
0 1 1 0 2
0 0 0 0 0
```

3. Total number of labeled objects = 2